

The Date Display GUI was designed using the MVC (Model-View-Controller) architectural pattern, where `DateModel` manages the date data, `DateView` handles the graphical interface, and `DateController` coordinates between them.

This separation ensures that each layer can change independently without affecting the others.

Main functionalities include displaying the current date in Vietnamese format (Ngày dd tháng MM năm yyyy), moving the date forward or backward by a given number of days or months, and validating user input with appropriate error messages for empty, non-numeric, negative, or excessively large values. Date edge cases such as leap years, month-end clamping (e.g. January 31 plus one month becomes February 28), and out of range dates are handled correctly through `java.time.LocalDate`.

The implementation uses Java Swing for the interface. `DateModel.java` stores the current `LocalDate` and notifies listeners via `PropertyChangeSupport` when the date changes. `DateView.java` defines all UI components and exposes a minimal public interface (`updateDate`, `showError`, `getInputText`, and listener registration methods) so that no business logic lives in the view. `DateController.java` validates input, calls the appropriate model method, and handles any `DateTimeException` that arises. `Main.java` wires the three layers together and launches the application on the Event Dispatch Thread using `SwingUtilities.invokeLater()`.